1 Introduction

2×2 matrices A, B have commutator $[A, B] = AB - BA$, and the **iterated** left-normed commutator of k matrices is $[A_1, \ldots, A_k]$ $[\ldots [[A_1, A_2], A_3], \ldots, A_k]$, matching the standard lower-central-series convention for Lie algebras. This object arose out of a question in algebraic complexity theory – how many scalar multiplications are needed to **compute** $[A_1, \ldots, A_k]$? – which remains genuinely open for $k \geq 3$ and is treated in full in the companion note k-iterated-commutators-complexity.typ. The one fact from that story needed here is purely algebraic, not complexity-theoretic: an explicit bilinear formula for the ordinary 2×2 commutator, chained across k matrices, gives an explicit polynomial parametrization of $[A_1, \ldots, A_k]$'s reduced coordinates (**??** 3.2 below), and it is the **vanishing locus** of this parametrization, as an algebraic variety and its coordinate ring, that is the actual subject of this article. (One naming note: the filename this article is kept under, ring-of-k-commuting-2x2-matrices, uses "commuting" in this generalized iterated-commutator-vanishing sense – $[A_1, \ldots, A_k] = 0$ – not literally "every pair of $A_1, \ldots, A_k$ commutes," a weaker condition than the classical pairwise commuting variety it specializes to at $k = 2$.)

The classical commuting variety. For A, B arbitrary (not necessarily 2×2) $n \times n$ matrices, the variety C_n $\{(A, B) \in M_n \times M_n : AB = BA\}$ – the ordinary, unreduced, un-iterated case this article's own $k = 2$ specializes to – has been studied since the 1950s. Its irreducibility was proven independently by Motzkin and Taussky MotzkinTaussky1955 and by Gerstenhaber Gerstenhaber1961, the latter also establishing the now-standard bound $\dim_{\mathbb{F}} \mathbb{F}[A, B] \leq n$ for the subalgebra generated by two commuting matrices. Whether the coordinate ring $\mathbb{F}[A, B]/I$, I the ideal generated by the entries of $AB - BA$, is **reduced** and **Cohen-Macaulay** for every n is a much harder question, attributed to Michael Artin and Melvin Hochster – an unpublished, informally-circulated conjecture with no single dateable original source ArtinHochsterConjecture, stated precisely as "Conjecture 1.1" in Majidi-Zolbanin and Snapp MajidiZolbaninSnapp. (A closely related, more specifically dated 1984 reducedness conjecture, for a different but adjacent ideal built from a commutator matrix's diagonal/anti-diagonal entries, is attributed to Hochster alone by Chau Chau2024.) Freyja Hreinsdóttir's body of work Hreinsdottir2005 – building on her own 1997 Stockholm doctoral thesis – attacks the Artin-Hochster conjecture case by case at $n = 3, 4$: syzygies, Betti numbers, the canonical module, and a Koszul-dual structure theorem, verified computationally through $n = 7$ but proven in general only in these small cases. Knutson Knutson2003 studies the minimal primes of closely related schemes from a different angle. Cohen-Macaulayness of the **classical** commuting variety therefore remains open, for every $n \geq 3$, at the time of writing.

At $n = 2$ specifically – this article's $k = 2$ base case – the question is comparatively easy, and is in effect answered again, as a special case, by **??** 6.2 below: I_2 (**??** 3.2) is literally the ideal of the $n = 2$ commuting variety, expressed in reduced (trace-removed) coordinates rather than the full matrix-entry coordinates the classical literature typically uses. The genuinely new object in this article is the **iterated** generalization to $k \geq 3$: not the classical commuting variety at all (see the naming note above), but a structurally analogous determinantal ideal, built by chaining the same bilinear cross-product formula that governs the $n = 2$ case. It is exactly the same **flavor** of question the Artin-Hochster conjecture asks – is this natural, geometrically meaningful ideal Cohen-Macaulay? is it reduced? – for a different, recursively-defined family of ideals, and this article answers it completely and unconditionally, for every $k \geq 2$ (**??** 8.1), rather than case by case or conjecturally. In that sense the results below can be read as a fully worked structural analogue of the still-open classical question, in a setting simple enough ($n = 2$, but k unbounded) to admit a complete answer.

2 Notation

Notation used across more than one later result, in order of first appearance; symbols local to a single proof are omitted.

A typographic convention, used throughout: an **overline** denotes a **vector** – an element of $\mathbb{F}^3$ (or, for R_k itself, of the free module R_k^3) – while the **same letter, plain, with a two-index comma subscript**, denotes one of that vector's three individual scalar **components**. So $\overline{v}_i \in \mathbb{F}^3$ is block i's whole reduced-coordinate vector, while its three components are the plain scalars $v_{i,1}, v_{i,2}, v_{i,3} \in \mathbb{F}$ (equivalently $v_{i,j}$ for $j = 1, 2, 3$) – never overlined, since a single scalar entry is not itself a vector. The same convention applies to $\overline{w}_j$ versus its components $w_{j,1}, w_{j,2}, w_{j,3}$.

Symbol	Meaning
$\overline{v}$ (also $\overline{v}_i$; components $v_{i,1}, v_{i,2}, v_{i,3}$, or $v_{i,j}$)	reduced-coordinate vector in $\mathbb{F}^3$, and its blocks
τ (and its matrix T)	the reshuffle $\tau(a, b, c)$ $(b, a, 2c)$
$\mathbb{F}$	ambient field, characteristic $\neq 2$
R_k	the polynomial ring $\mathbb{F}[\overline{v}_1, \ldots, \overline{v}_k]$
I_k	the defining ideal, $I_k \subset R_k$
$\overline{w}_j$	(a) the recursive vector in $\mathbb{F}^3$; (b) as a polynomial map $\mathbb{A}^{3j} \to \mathbb{A}$
U_j	the Zariski-open locus where $\overline{w}_j \neq 0$
π	the projection $\mathbb{A}^{3k} \to \mathbb{A}^{3(k-1)}$ forgetting $\overline{v}_k$
Φ	the map $\mathbb{A}^{3(k-1)} \times \mathbb{A}^1 \to \mathbb{A}^{3k}$
Y_1, Y_2	the two strata of $V(I_k)$
P_j $(j = 2, \ldots, k)$	the primary components of I_k
J_k	the Jacobian of $\overline{w}_k$
$[u]_\times$	skew matrix with $[u]_\times v = u \times v$

3 Setup: reduced coordinates and the defining ideal

Step 1: discard the scalar part. Write $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ as the sum of its scalar part $(\mathrm{tr}(A)/2)\, I$ and its traceless remainder A_0 $A - (\mathrm{tr}(A)/2)\, I = \begin{pmatrix} d_a/2 & a_{12} \\ a_{21} & -d_a/2 \end{pmatrix}$, $d_a\, a_{11} - a_{22}$. Scalar matrices are central, so they never contribute to a commutator: $[cI, B] = cIB - BcI = cB - cB = 0$ for every B and every scalar c, hence $[A, B] = [A_0, B_0]$ identically – no information relevant to **any** bracket is lost by discarding the scalar part, so from here on only the traceless remainder matters.

Step 2: encode the traceless remainder as a triple. A traceless matrix $\begin{pmatrix} x & y \\ z & -x \end{pmatrix}$ has exactly three free entries – the $(2, 2)$ entry is forced once x, y, z are fixed – giving a bijection onto $\mathbb{R}^3$. Write

$$\mathrm{red}(A)\ (a_{12}, a_{21}, d_a) \in \mathbb{R}^3$$

for this encoding (so $\mathrm{red}(A) = \mathrm{red}(A_0)$, reading A's traceless part off directly without subtracting the scalar part by hand first) – this is the notation $\overline{v}$ used throughout the rest of the article. Matching Step 1's $A_0 = \begin{pmatrix} d_a/2 & a_{12} \\ a_{21} & -d_a/2 \end{pmatrix}$: red's three coordinates are literally A_0's free entries, except that d_a stands in for **twice** the $(1, 1)$-entry rather than the entry itself – a convention fixed once and for all here purely so that Step 4 below comes out with no stray constants.

Step 3: form the commutator. Given traceless $A_0 = \begin{pmatrix} x & y \\ z & -x \end{pmatrix}$ and $B_0 = \begin{pmatrix} u & v \\ w & -u \end{pmatrix}$, the ordinary matrix commutator $[A_0, B_0] = A_0B_0 - B_0A_0$ is again traceless automatically ($\mathrm{tr}(PQ) = \mathrm{tr}(QP)$ for any P, Q, so $\mathrm{tr}([P, Q]) = 0$ always), hence again encodable by red.

Step 4: the encoded bracket is the cross product – up to one fixed reshuffle. Carrying out Step 3's multiplication in coordinates (the $\mathfrak{sl}_2(\mathbb{C}) \cong \mathfrak{so}(3) \cong (\mathbb{R}^3, \times)$ isomorphism made explicit) and writing $[A_0, B_0] = \begin{pmatrix} x' & y' \\ z' & -x' \end{pmatrix}$, one checks directly that, with $\mathrm{red}(A)$ and $\mathrm{red}(B)$ standing in for themselves (**not** abbreviated to bare letters here, to avoid colliding with Step 3's plain scalars u, v, w),

$$z' = (\mathrm{red}(A) \times \mathrm{red}(B))_1\,, \quad y' = (\mathrm{red}(A) \times \mathrm{red}(B))_2\,, \quad x' = (\mathrm{red}(A) \times \mathrm{red}(B))_3.$$

Reading these three equalities off against $\mathrm{red}([A, B]) = (y', z', 2x')$ (Step 2's convention, applied to $[A, B]$ itself) gives

$$\mathrm{red}([A, B]) = \tau(\mathrm{red}(A) \times \mathrm{red}(B)), \quad \tau(a, b, c) \ (b, a, 2c).$$

The right-hand side is **not** the raw cross product $\mathrm{red}(A) \times \mathrm{red}(B)$ itself, but that cross product composed with the fixed invertible reshuffle τ (invertible whenever $2 \neq 0$): the $(1, 2)$-swap comes from red's coordinate order (a_{12} before a_{21}, the transpose of the naive (x, y, z) order), and the factor of 2 on the third slot comes from Step 2's $d_a = 2x$ convention. Both are bookkeeping artifacts of the coordinate choice fixed in Step 2 – τ is already forced at this single-bracket level, not a new phenomenon introduced only once brackets start chaining below.

Remark 3.1: why ?? 3.2's $\overline{w}_j$ carries no outer τ

?? 3.2 below sets $\overline{w}_2\ \overline{v}_1 \times \overline{v}_2$ – the **untwisted** cross product, not $\mathrm{red}([A, B]) = \tau(\overline{v}_1 \times \overline{v}_2)$ itself as Step 4 above would suggest. This is deliberate: for any fixed invertible linear map τ and any polynomial vector $f = (f_1, f_2, f_3)$, the ideal generated by the components of $\tau(f)$ equals the ideal generated by the components of f – each component of one is an $\mathbb{F}$-linear combination of the components of the other (via τ, respectively τ^{-1}), so the two generating sets differ only by an invertible change of generators, and invertible changes of generators do not change the ideal generated. Hence $I_2\ (w_{2,1}, w_{2,2}, w_{2,3})$ already equals the ideal generated by $\mathrm{red}([A, B])$'s three components, with no need to carry the extra τ explicitly – $\overline{w}_j$ is used purely because it is the cheaper quantity to compute recursively (one cross product per step, not a cross product composed with τ at every stage), and the single τ that **does** appear in **??** 3.2's recursion below is exactly the one deferred from the previous step, not a new one introduced fresh at each stage.

Chaining this bilinear formula across k matrices – each new bracket stage reuses the previous stage's output coordinates directly, no new input data needed – requires tracking how one stage's raw cross-product output is reinterpreted as the next stage's input vector: precisely the τ just discussed, applied once per new stage. This gives the following exact recursive description:

Definition 3.2: reduced iterated commutator

$$\overline{w}_2\ \overline{v}_1 \times \overline{v}_2, \quad \overline{w}_j\ \tau(\overline{w}_{j-1}) \times \overline{v}_j \quad (j = 3, \ldots, k).$$

Let $R_k\ \mathbb{Q}[\overline{v}_1, \ldots, \overline{v}_k]$ ($\overline{v}_i \in \mathbb{Q}^3$, $3k$ variables) and $I_k \subset R_k$ the ideal generated by the three components of $\overline{w}_k$ – the vanishing ideal of $[A_1, \ldots, A_k] = 0$ in reduced coordinates.

To spell out exactly what this means: "$R_k\ \mathbb{Q}[\overline{v}_1, \ldots, \overline{v}_k]$" is shorthand for **the polynomial ring in the $3k$ scalar variables $v_{i,j}$ ($i = 1, \ldots, k$, $j = 1, 2, 3$)**, grouped into k blocks of 3 –

not a ring with k vector-valued generators (a polynomial ring's generators are always scalars). At $k = 2$, written out in full:

$$R_2 = \mathbb{Q}\left[v_{1,1}, v_{1,2}, v_{1,3}, v_{2,1}, v_{2,2}, v_{2,3}\right],$$

six ordinary scalar indeterminates, with $\overline{v}_1 = (v_{1,1}, v_{1,2}, v_{1,3})$ and $\overline{v}_2 = (v_{2,1}, v_{2,2}, v_{2,3})$ merely a notational grouping of them into two blocks of three – convenient because $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ is most naturally written as a cross product of two block-vectors, not as some formula in six unrelated scalars.

Each $\overline{w}_j$ is a triple of polynomials, i.e. $\overline{w}_j \in R_j^3$ (the free rank-3 module over R_j) – equivalently, exactly as recorded in the Notation table's entry (b) for $\overline{w}_j$, the polynomial map $\overline{w}_j : \mathbb{A}^{3j} \to \mathbb{A}^3$ **given by substitution**: evaluating each of $\overline{w}_j$'s three polynomial components at a point $(\overline{v}_1, \ldots, \overline{v}_j) \in \mathbb{A}^{3j}$. Concretely at $k = 2$, writing out all six variables, $\overline{w}_2$ is the substitution map

$$\begin{pmatrix} v_{1,1} \\ v_{1,2} \\ v_{1,3} \\ v_{2,1} \\ v_{2,2} \\ v_{2,3} \end{pmatrix} \in \mathbb{A}^6 \quad \mapsto \quad \begin{pmatrix} v_{1,2}v_{2,3} - v_{1,3}v_{2,2} \\ v_{1,3}v_{2,1} - v_{1,1}v_{2,3} \\ v_{1,1}v_{2,2} - v_{1,2}v_{2,1} \end{pmatrix} \in \mathbb{A}^3,$$

and I_k (**??** 3.2) is, equivalently, the ideal generated by the three coordinate functions of $\overline{w}_k : \mathbb{A}^{3k} \to \mathbb{A}^3$ pulled back along this substitution.

Example 3.3: $\overline{w}_k$ and I_k, explicitly, for $k = 2, 3, 4$

Writing $\overline{v}_i = (v_{i,1}, v_{i,2}, v_{i,3})$ for the three coordinates of block $\overline{v}_i$: at $k = 2$, the three components of $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ are

$$v_{1,2}v_{2,3} - v_{1,3}v_{2,2}, \quad v_{1,3}v_{2,1} - v_{1,1}v_{2,3}, \quad v_{1,1}v_{2,2} - v_{1,2}v_{2,1},$$

and I_2 $(w_{2,1}, w_{2,2}, w_{2,3})$ is the ideal they generate – $\overline{w}_2$ is simply the vector whose components those three polynomials are (**??** 3.1 explains why it is $\overline{w}_2$, not $\tau(\overline{w}_2) = \operatorname{red}([A_1, A_2])$ itself, that plays this role). At $k = 3$, the three components of $\overline{w}_3 = \tau(\overline{w}_2) \times \overline{v}_3$ are

$$2v_{1,2}v_{2,1}v_{3,2} - 2v_{1,1}v_{2,2}v_{3,2} - v_{1,3}v_{2,2}v_{3,3} + v_{1,2}v_{2,3}v_{3,3},$$

$$-2v_{1,2}v_{2,1}v_{3,1} + 2v_{1,1}v_{2,2}v_{3,1} - v_{1,3}v_{2,1}v_{3,3} + v_{1,1}v_{2,3}v_{3,3},$$

$$v_{1,3}v_{2,2}v_{3,1} - v_{1,2}v_{2,3}v_{3,1} + v_{1,3}v_{2,1}v_{3,2} - v_{1,1}v_{2,3}v_{3,2},$$

and I_3 $(w_{3,1}, w_{3,2}, w_{3,3})$ likewise. At $k = 4$, $\overline{w}_4 = \tau(\overline{w}_3) \times \overline{v}_4$: the same pattern continues, now quartic with up to 8 terms per component – e.g. the first component of $\overline{w}_4$ is

$$-2v_{1,3}v_{2,2}v_{3,1}v_{4,2} + 2v_{1,2}v_{2,3}v_{3,1}v_{4,2} - 2v_{1,3}v_{2,1}v_{3,2}v_{4,2} + 2v_{1,1}v_{2,3}v_{3,2}v_{4,2}$$

$$+2v_{1,2}v_{2,1}v_{3,2}v_{4,3} - 2v_{1,1}v_{2,2}v_{3,2}v_{4,3} - v_{1,3}v_{2,2}v_{3,3}v_{4,3} + v_{1,2}v_{2,3}v_{3,3}v_{4,3},$$

the other two of the same shape (omitted here for space; all three components of $\overline{w}_4$, and I_4 $(w_{4,1}, w_{4,2}, w_{4,3})$, computed and verified in code/verify_determinantal_structure_all_k.sage).

Stated over $\mathbb{Q}$ since that is what the computational checks below actually run over, but nothing in the construction uses $\mathbb{Q}$ specifically: every result in this article is proven for R_k $\mathbb{F}\left[\overline{v}_1, \ldots, \overline{v}_k\right]$ over an **arbitrary** field $\mathbb{F}$ of characteristic $\neq 2$ (needed only for τ's invertibility), by the identical argument with $\mathbb{Q}$ replaced by $\mathbb{F}$ throughout.

4 The determinantal structure and the generator count

Theorem 4.1: determinantal structure

For $k \geq 3$, I_k is exactly the ideal of 2×2 minors of the 2×3 matrix

$$\begin{pmatrix} \tau(\overline{w}_{k-1}) \\ \overline{v}_k \end{pmatrix},$$

i.e. $V(I_k)$ is the preimage, under $(\overline{v}_1, \ldots, \overline{v}_k) \mapsto \begin{pmatrix} \tau(\overline{w}_{k-1}(\overline{v}_1, \ldots, \overline{v}_{k-1})) \\ \overline{v}_k \end{pmatrix}$, of the classical rank-$\leq 1$ determinantal variety of 2×3 matrices. (At $k = 2$ the same statement holds with $\tau(\overline{w}_{k-1})$ replaced by $\overline{v}_1$ itself, matching **??** 3.2's base case $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ directly.)

Proof of **??** *3.3.* Immediate from the fact that the three components of a cross product $a \times b$ are (up to sign) exactly the 2×2 minors of the 2×3 matrix with rows a, b, combined with **??** 3.2's recursion – and checked independently: comparing an independent implementation of the τ/cross-product recursion against the ideal actually produced by the chained bilinear formula gives the **exact same ideal** (Gröbner-basis containment both ways) at $k = 2, \ldots, 6$ (code/verify_determinantal_structure_all_k.sage); the underlying polynomial identity $\operatorname{red}([A, B]) = \tau(\operatorname{red}(A) \times \operatorname{red}(B))$ makes this a two-line induction valid for **every** k, not merely a pattern checked at small k (code/verify_tau_cross_identity.sage). □

This theorem is the engine behind the geometric results of §5: R_k/I_k is not merely **determinantal-looking**, it is literally an iterated extension of the coordinate ring of a classical determinantal variety.

Example 4.2: the 2×3 matrix of ?? 3.3, for $k = 2, 3, 4$

At $k = 2$:

$$\begin{pmatrix} \overline{v}_1 \\ \overline{v}_2 \end{pmatrix} = \begin{pmatrix} v_{1,1} & v_{1,2} & v_{1,3} \\ v_{2,1} & v_{2,2} & v_{2,3} \end{pmatrix}.$$

At $k = 3$, the first row is $\tau(\overline{w}_2) = (w_{2,2}, w_{2,1}, 2w_{2,3})$ where $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ is as in the previous example, giving

$$\begin{pmatrix} \tau(\overline{w}_2) \\ \overline{v}_3 \end{pmatrix} = \begin{pmatrix} v_{1,3}v_{2,1} - v_{1,1}v_{2,3} & -v_{1,3}v_{2,2} + v_{1,2}v_{2,3} & -2v_{1,2}v_{2,1} + 2v_{1,1}v_{2,2} \\ v_{3,1} & v_{3,2} & v_{3,3} \end{pmatrix}.$$

At $k = 4$, the first row is $\tau(\overline{w}_3)$, entries cubic with up to 4 terms:

$$\begin{pmatrix} -2v_{1,2}v_{2,1}v_{3,1} + 2v_{1,1}v_{2,2}v_{3,1} - v_{1,3}v_{2,1}v_{3,3} + v_{1,1}v_{2,3}v_{3,3} & \cdots & \cdots \\ v_{4,1} & v_{4,2} & v_{4,3} \end{pmatrix}$$

(the other two entries of the first row, of the same shape, omitted for width – all displayed in full in code/verify_determinantal_structure_all_k.sage). The row degree grows by one at each step $(1, 2, 3, \ldots)$, while the second row is always the fresh, free vector $\overline{v}_k$ – exactly the "one fixed complicated row, one free row" shape **??** 5.15 needs.

Lemma 4.3: non-degeneracy

For every $j \geq 2$, the polynomial map $\overline{w}_j : \mathbb{A}^{3j} \to \mathbb{A}^3$ is not identically zero. Hence $U_j \ \{p \in \mathbb{A}^{3j} : \overline{w}_j(p) \neq 0\}$ is a nonempty Zariski-open subset.

Proof of **??** *4.2.* By induction: $\overline{w}_2(e_1, e_2) = e_1 \times e_2 = e_3 \neq 0$; given $c\ \overline{w}_{j-1}(p) \neq 0$, at least one standard basis vector is not proportional to $\tau(c)$, and a cross product vanishes only on linearly dependent pairs, so $\overline{w}_{j(p,e_i)} = \tau(c) \times e_i \neq 0$ for a suitable e_i.

□

The generator count of I_k needs nothing beyond the definition and non-degeneracy – not codimension, not the determinantal-structure theorem above, not any external citation:

Lemma 4.4: independent same-degree generators are minimal

If a homogeneous ideal in a graded ring is generated by finitely many forms of the same degree that are linearly independent over the base field, that generating set is minimal.

Proof of **??** *4.4.* A polynomial-coefficient combination reproducing one of the generators from the others, while staying in the same degree, can only use degree-0 (constant) coefficients – so a redundant generator would violate linear independence directly.

□

Lemma 4.5: the three generators of I_k are independent

The three components of $\overline{w}_k$ are linearly independent over $\mathbb{F}$, for every $k \geq 2$.

Proof of **??** *4.5.* Checked directly (coefficient-matrix rank 3) at $k = 2, \ldots, 6$. Structurally: a nontrivial constant linear relation among the components of $\overline{w}_k = \tau(\overline{w}_{k-1}) \times \overline{v}_k$ would force $\tau(\overline{w}_{k-1})$ itself to be a constant vector, which it is not for $k \geq 3$ (**??** 4.2's non-degeneracy, τ invertible) nor for $k = 2$ ($\overline{v}_1$ is a free variable block, not a constant).

□

Theorem 4.6: I_k has exactly 3 minimal generators

For every $k \geq 2$: I_k has exactly 3 minimal generators.

Proof of **??** *4.6.* Immediate from **??** 4.4 applied to **??** 4.5's independence, since I_k is generated by the three (same-degree-k-homogeneous) components of $\overline{w}_k$ by definition.

□

5 The geometric structure of $V(I_k)$

Lemma 5.1: continuous images and closures preserve irreducibility

Let $f : X \to Y$ be a continuous map of topological spaces with X irreducible. Then $f(X)$ (with the subspace topology) is irreducible, and hence so is $\overline{f(X)}$.

Proof of **??** *5.1.* **Image.** Suppose $f(X) = C_1 \cup C_2$ with C_1, C_2 closed in $f(X)$. By continuity, $f^{-1}(C_1)$ and $f^{-1}(C_2)$ are closed in X, and $X = f^{-1}(C_1) \cup f^{-1}(C_2)$. Since X is irreducible, one of the two equals X – say $f^{-1}(C_1) = X$ – so $f(X) \subseteq C_1$, i.e. $C_1 = f(X)$, not proper. Hence $f(X)$ is irreducible. **Closure.** If $A \subseteq Y$ is irreducible and $\overline{A} = C_1 \cup C_2$ with C_1, C_2 closed in Y, then $A = (A \cap C_1) \cup (A \cap C_2)$; irreducibility of A forces, say, $A \subseteq C_1$; since C_1 is closed, $\overline{A} \subseteq C_1$, and combined with $C_1 \subseteq \overline{A}$, $C_1 = \overline{A}$, not proper. Apply both parts with A $f(X)$.

□

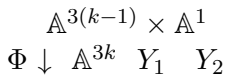


Figure 1: Both strata of $V(I_k) = Y_1 \cup Y_2$ inside the same ambient $\mathbb{A}^{3k}$: Y_1 $V(I_{k-1}) \times \mathbb{A}^3$ (identifying $\mathbb{A}^{3(k-1)} \times \mathbb{A}^3 \cong \mathbb{A}^{3k}$) and Y_2 $\overline{\Phi(\mathbb{A}^{3k-2})}$, Φ's image (Φ shown mapping in from above, into Y_2 specifically – Y_1 is not part of Φ's image). The two overlap (not disjoint, Propositions 17-18 rule out either containing the other): $\Phi(p, 0) = (p, 0)$ for **every** p, so the whole zero-section $\{(p, 0)\}$ lies in Y_2, and its intersection with Y_1 is $V(I_{k-1}) \times \{0\}$, of dimension $3k - 5$ – properly smaller than either stratum's own dimension $3k - 2$.

This is purely topological – continuity of f is the only hypothesis used, nothing about f being a polynomial map or morphism of varieties Youcis2014 (working-notes/ topological-irreducibility-of-image.org has the fuller discussion, including why this observation matters for a possible future Lean formalization of this article).

Definition 5.2: the strata Y_1, Y_2

For $k \geq 3$, write π for the projection **forgetting** $\overline{v}_k$,

$$\pi : \mathbb{A}^{3k} \to \mathbb{A}^{3(k-1)}, \quad \pi(\overline{v}_1, \ldots, \overline{v}_k)\ (\overline{v}_1, \ldots, \overline{v}_{k-1}),$$

and Φ for the map

$$\Phi : \mathbb{A}^{3(k-1)} \times \mathbb{A}^1 \to \mathbb{A}^{3k}, \quad \Phi(p, \lambda) \quad \begin{matrix} (p, \\ \lambda\tau(\overline{w}_{k-1}(p))) \end{matrix}.$$

Define two subsets of $\mathbb{A}^{3k}$:

$$Y_1\ V(I_{k-1}) \times \mathbb{A}^3, \quad Y_2\ \overline{\Phi(\mathbb{A}^{3k-2})}.$$

Y_1 is the locus where the previous bracket already vanishes, with $\overline{v}_k$ completely free; Y_2 is the closure of the locus swept out as $\overline{v}_k$ ranges over scalar multiples of $\tau(\overline{w}_{k-1}(p))$, over all $p \in \mathbb{A}^{3(k-1)}$. At $k = 2$ there is no "previous bracket" to chain from, so neither Y_1 nor Φ is defined; by convention Y_2 $V(I_2)$ directly there, matching **??** 3.3's own base-case treatment.

Lemma 5.3: Y_2 is irreducible

For $k \geq 3$: Y_2 (**??** 5.2) is irreducible, and in fact its defining ideal (the graph ideal of Φ with λ eliminated) is prime.

Proof of **??** *5.3.* Φ is a polynomial map, hence continuous for the Zariski topology (preimages of Zariski-closed sets under a polynomial map are again Zariski-closed, by substitution), and its domain $\mathbb{A}^{3k-2}$ is irreducible (the polynomial ring over $\mathbb{F}$ is a domain, so the zero ideal is prime). **??** 5.1 gives Y_2 irreducible directly. Primality of the elimination ideal follows the same way: it is the kernel of a map into a domain ($\mathbb{F}[\overline{v}_1, \ldots, \overline{v}_k]/(\cdot) \cong \mathbb{F}[\overline{v}_1, \ldots, \overline{v}_{k-1}, \lambda]$), hence prime – this holds for every k by this argument alone, additionally corroborated computationally at $k = 4$ by computing the elimination ideal directly and confirming primality in Sage, as an independent check rather than a substitute for the argument.

□

Example 5.4: the defining ideals of Y_1 and Y_2, for $k = 2, 3, 4$

Writing $\overline{v}_i = (v_{i,1}, v_{i,2}, v_{i,3})$ for the three coordinates of block $\overline{v}_i$: at $k = 2$, Y_2 $V(I_2)$ (the base-case convention above), so its ideal **is** I_2, generated by

$$v_{1,2}v_{2,3} - v_{1,3}v_{2,2}, \quad v_{1,3}v_{2,1} - v_{1,1}v_{2,3}, \quad v_{1,1}v_{2,2} - v_{1,2}v_{2,1}.$$

(Y_1 is not defined at $k = 2$, **??** 5.2.) For $k \geq 3$, Y_1's defining ideal is simply I_{k-1} itself, viewed inside R_k by leaving $\overline{v}_k$ free – at $k = 3$, exactly the three generators displayed above for I_2, now inside R_3 with $\overline{v}_3$ not appearing at all; at $k = 4$, exactly I_3's three generators (displayed below in this same example) inside R_4 with $\overline{v}_4$ free. Unlike Y_2, no elimination is needed to compute Y_1's ideal – it is already I_{k-1}'s own generating set, unchanged. At $k = 3$, Y_2's elimination ideal (computed directly, not inferred) is generated by

$$2v_{2,2}v_{3,1} + 2v_{2,1}v_{3,2} + v_{2,3}v_{3,3}, \quad 2v_{1,2}v_{3,1} + 2v_{1,1}v_{3,2} + v_{1,3}v_{3,3},$$

$$2v_{1,2}v_{2,1}v_{3,2} - 2v_{1,1}v_{2,2}v_{3,2} - v_{1,3}v_{2,2}v_{3,3} + v_{1,2}v_{2,3}v_{3,3}$$

– notably, two of the three generators are **linear** in $\overline{v}_3$ (exactly the two components of the "proportionality" condition $\overline{v}_3 \propto \tau(\overline{w}_2)$), and only the third, not involving $\overline{v}_3$ at all, carries the full complexity of $\tau(\overline{w}_2)$ itself. **This ideal is a proper factor of I_3, not equal to it** – I_3 is the intersection of this ideal with a second prime (the "Y_1" piece, where the $k = 2$ bracket already vanishes); see the component-count example after **??** 5.11 below, where this exact ideal reappears as one of I_3's two primary components. At $k = 4$, by the same pattern, two of the three generators are linear in $\overline{v}_4$ and the third is not, now degree 4:

$$2v_{3,2}v_{4,1} + 2v_{3,1}v_{4,2} + v_{3,3}v_{4,3},$$

$$v_{1,3}v_{2,2}v_{4,1} - v_{1,2}v_{2,3}v_{4,1} - v_{1,3}v_{2,1}v_{4,2} + v_{1,1}v_{2,3}v_{4,2} + v_{1,2}v_{2,1}v_{4,3} - v_{1,1}v_{2,2}v_{4,3},$$

$$2v_{1,3}v_{2,2}v_{3,1}v_{4,2} - 2v_{1,2}v_{2,3}v_{3,1}v_{4,2} + 2v_{1,3}v_{2,1}v_{3,2}v_{4,2} - 2v_{1,1}v_{2,3}v_{3,2}v_{4,2}$$

$$-2v_{1,2}v_{2,1}v_{3,2}v_{4,3} + 2v_{1,1}v_{2,2}v_{3,2}v_{4,3} + v_{1,3}v_{2,2}v_{3,3}v_{4,3} - v_{1,2}v_{2,3}v_{3,3}v_{4,3}.$$

All computed directly (elimination of λ from the graph ideal of Φ, in Sage), not just asserted from the pattern.

Remark 5.5: the closure, not the raw image, is what is needed

A polynomial map's image need not itself be closed – the standard example is $\mathbb{A}^2 \to \mathbb{A}^2$, $(x, y) \mapsto (x, xy)$, whose image is not even locally closed, though it is **constructible** (a finite union of locally closed sets) by Chevalley's theorem, which has no purely topological analogue for general continuous maps Youcis2014. This never causes a problem for Y_2, because Y_2 is **defined** as $\overline{\Phi(\cdot)}$ from the outset, and the decomposition used in **??** 5.7's proof below, $V(I_k) = Y_1 \cup Y_2$, is established by a direct case split rather than by taking closures and hoping: a point $(p, \overline{v}_k) \in V(I_k)$ either has $\tau(\overline{w}_{k-1}(p)) = 0$, putting it in Y_1 directly, or has $\overline{v}_k = \lambda\tau(\overline{w}_{k-1}(p))$ for some λ, so $(p, \overline{v}_k) = \Phi(p, \lambda)$ is **literally** a value of Φ – not merely a limit point – hence in Y_2 with no appeal to closure at all. The reverse containment $Y_1 \cup Y_2 \subseteq V(I_k)$ needs only that $\Phi(\cdot) \subseteq V(I_k)$ (a cross product of a vector with itself vanishes: $\tau(\overline{w}_{k-1}) \times (\lambda\tau(\overline{w}_{k-1})) = 0$ identically) together with $V(I_k)$ already being closed, automatic since it is a vanishing locus by definition. So the theorem never needs "the image of Φ is closed" – only that it is contained in the already-closed $V(I_k)$, which is immediate.

Lemma 5.6: dimension of Y_2, and where it lives

With Y_2 as in **??** 5.3 and U_{k-1} as in **??** 4.2: $\dim Y_2 = 3k-2$, and Y_2 meets $\pi^{-1}(U_{k-1})$ nonemptily, where π is the projection of **??** 5.2, forgetting $\overline{v}_k$.

Proof of **??** *5.6.* $\pi : \mathbb{A}^{3k} \to \mathbb{A}^{3(k-1)}$ discards the last block: $\pi(\overline{v}_1, \ldots, \overline{v}_k) = (\overline{v}_1, \ldots, \overline{v}_{k-1})$. Hence $\pi^{-1}(U_{k-1}) = U_{k-1} \times \mathbb{A}^3$, the set of points whose **first** $k-1$ blocks lie in U_{k-1} and whose $\overline{v}_k$ is completely unconstrained. Over U_{k-1}, Φ is injective. Both p and λ can be recovered from $\Phi(p, \lambda)$: p directly, and λ by dividing any nonzero coordinate of $\overline{v}_k = \lambda\tau(\overline{w}_{k-1}(p))$ by the corresponding coordinate of $\tau(\overline{w}_{k-1}(p))$; this division is well-defined since $\tau(\overline{w}_{k-1}(p)) \neq 0$ on U_{k-1}. Injectivity gives $\dim Y_2 = \dim\left(U_{k-1} \times \mathbb{A}^1\right) = 3(k-1)+1 = 3k-2$. Since U_{k-1} is nonempty by **??** 4.2, $\Phi(U_{k-1} \times \{1\})$ is a nonempty subset of $Y_2 \cap \pi^{-1}(U_{k-1})$.

□

Theorem 5.7: I_k has codimension 2

For every $k \geq 2$: $\dim V(I_k) = 3k-2$, i.e. $\operatorname{codim}(I_k) = 2$.

Proof of **??** *5.7.* Induction on k. **Base case** $k = 2$: $V(I_2)$ is the rank-≤ 1 locus of a 2×3 matrix with independent-indeterminate entries, of the standard dimension $4 = 3 \cdot 2 - 2$ for the generic determinantal variety of this shape MotzkinTaussky1955. **Inductive step**: by **??** 3.3, and since $a \times b = 0$ if and only if a, b are linearly dependent, $V(I_k) = Y_1 \cup Y_2$ (**??** 5.2). The locus $\tau(\overline{w}_{k-1}) = 0$ gives Y_1, with $\overline{v}_k$ unconstrained; the locus $\tau(\overline{w}_{k-1}) \neq 0$ with $\overline{v}_k$ proportional to it gives a subset of Y_2. This is the direct case split of **??** 5.4, not merely a closure-level approximation. By the inductive hypothesis, $\dim V(I_{k-1}) = 3(k-1)-2$, so $\dim Y_1 = 3(k-1)-2+3 = 3k-2$. Also $\dim Y_2 = 3k-2$, by **??** 5.6. Hence $\dim V(I_k) = \dim(Y_1 \cup Y_2) = 3k-2$. This dimension statement alone needs no containment argument; that is needed only for the sharper component-count result below.

□

Example 5.8: dimension and codimension, for $k = 2, 3, 4$

Values of $\dim R_k$ (the ambient dimension, $= 3k$), $\dim(R_k/I_k)$ (the dimension of the variety $V(I_k)$ – **not** "$\dim I_k$", an ideal is not a variety and does not itself have a dimension in this sense), and $\operatorname{codim}(I_k)$ $\dim R_k - \dim(R_k/I_k)$:

k	$\dim R_k$	$\dim(R_k/I_k)$	$\operatorname{codim}(I_k)$
2	6	4	2
3	9	7	2
4	12	10	2

All three rows follow directly from **??** 5.7's formula $\dim(R_k/I_k) = 3k-2$; also confirmed by direct Gröbner-basis dimension computation in Sage at each k, as an independent check.

Proposition 5.9: $Y_2 \not\subseteq Y_1$

With Y_1, Y_2 as in the proof of **??** 5.7: $Y_2 \not\subseteq Y_1$.

Proof of **??** *5.8.* Pick $p \in U_{k-1}$ (**??** 4.2); then $\Phi(p, 0) = (p, 0) \in Y_2$ but $\overline{w}_{k-1}(p) \neq 0$ puts $(p, 0) \notin Y_1$.

□

Proposition 5.10: no component of Y_1 lies in Y_2

No irreducible component Z of $V(I_{k-1})$ (each of dimension $3(k-1)-2$, by **??** 5.7 applied at $k-1$) has $Z \times \mathbb{A}^3 \subseteq Y_2$.

Proof of **??** *5.10.* Both $Z \times \mathbb{A}^3$ and Y_2 are irreducible: Y_2 by **??** 5.3, and $Z \times \mathbb{A}^3$ because a product of irreducible varieties is irreducible (Z is irreducible by hypothesis, and $\mathbb{A}^3$ trivially is). Both have the **same** dimension $3k-2$, by **??** 5.7 and **??** 5.6. Containment would therefore force equality. This is refuted: $Z \times \mathbb{A}^3$ is disjoint from $\pi^{-1}(U_{k-1})$, since $\overline{w}_{k-1}$ vanishes identically on $Z \subset V(I_{k-1})$; but Y_2 meets $\pi^{-1}(U_{k-1})$, by **??** 5.6.

□

Theorem 5.11: component count

For every $k \geq 2$: $V(I_k)$ has exactly $k-1$ irreducible components, each of dimension $3k-2$. The component indexed $j = 2, \ldots, k$ involves exactly the first j coordinate blocks and nothing later.

Proof of **??** *5.11.* Induction on k. **Base case** $k = 2$: the rank-≤ 1 locus of a generic 2×3 matrix is irreducible – the affine cone over the Segre embedding $\mathbb{P}^1 \times \mathbb{P}^2 \hookrightarrow \mathbb{P}^5$, equivalently the Motzkin-Taussky MotzkinTaussky1955 irreducibility of the 2×2 commuting variety – so exactly 1 component, matching $k-1=1$. **Step**: $V(I_k) = Y_1 \cup Y_2$ as in **??** 5.7's proof; by the inductive hypothesis Y_1 has $k-2$ irreducible components, each of dimension $3k-2$ (**??** 5.7), none contained in another; **??** 5.8 and **??** 5.10 rule out containment between Y_2 and any of them or vice versa. A finite union of irreducible closed sets with no containments is irredundant, giving exactly $(k-2)+1 = k-1$ components.

□

This gives I_k's minimal primes and their count, but not yet that I_k is radical (so not yet that these are literally the components of a **primary** decomposition of I_k itself, as opposed to of $\sqrt{I_k}$) – that is closed in §6, after Cohen-Macaulayness is available.

Definition 5.12: the components P_j

For $k \geq 2$ and $j = 2, \ldots, k$, write P_j for the vanishing ideal of the component of $V(I_k)$ indexed j in **??** 5.11 – the one involving exactly the first j coordinate blocks. Explicitly, recursively in j:

- **Base case,** $j = k$: P_k is Y_2's own defining ideal at parameter k (**??** 5.3).
- **Recursive case,** $j < k$: P_j is that same ideal computed at parameter j instead – i.e. Y_2's defining ideal when the construction is run with $k = j$ – viewed inside R_k by leaving $\overline{v}_{j+1}, \ldots, \overline{v}_k$ free.
- **Consequence**: P_j does not depend on which $k \geq j$ it is viewed inside of – it is the same ideal, just embedded in a ring with more or fewer free variables.

Example 5.13: a suitable point of Y_2, concretely, at $k = 3$

Rather than only the defining ideal (Example 24 below), a single explicit point: take $\overline{v}_1 = (1,0,0)$, $\overline{v}_2 = (0,1,0)$ (so $\overline{v}_1 \in U_2$, **??** 4.2), giving $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2 = (0,0,1)$ and

$\tau(\overline{w}_2) = (0,0,2)$. Taking $\lambda = 1$ in **??** 5.2's Φ, i.e. $\overline{v}_3\ \tau(\overline{w}_2) = (0,0,2)$, gives the point

$$(\overline{v}_1, \overline{v}_2, \overline{v}_3) = ((1,0,0),(0,1,0),(0,0,2)) \in Y_2 \subset \mathbb{A}^9.$$

Directly: $\overline{w}_3 = \tau(\overline{w}_2) \times \overline{v}_3 = (0,0,2) \times (0,0,2) = 0$ (a vector crossed with itself), confirming this point lies in $V(I_3)$, as any point of the form $\Phi(p,\lambda)$ must (**??** 5.2). This is also, concretely, a point of the largest-dimension inherited case: it is **not** a rank-2 witness in **??** 7.5's sense (that witness uses $\lambda = 0$, not $\lambda = 1$) – it illustrates Y_2 itself, not the radicality argument's own witness points.

Proposition 5.14: rational parametrization of each component

For $k \geq 2$ and $j = 2, \ldots, k$, the component of $V(I_k)$ indexed j (**??** 5.11) is a **rational** variety: it is the closure of the image of the polynomial map

$$\mathbb{A}^{3(j-1)} \times \mathbb{A}^1 \times \mathbb{A}^{3(k-j)} \to \mathbb{A}^{3k}, \quad (p,\lambda,q) \mapsto (\Phi_j(p,\lambda), q)\,,$$

where Φ_j is Φ (**??** 5.2) run at parameter j in **??** 5.12" sense (at $j = 2$, $\Phi_2(p,\lambda)$ $(p, \lambda p)$, matching **??** 3.3's own base-case convention), and q fills the remaining free blocks $\overline{v}_{j+1}, \ldots, \overline{v}_k$. This map is moreover **birational** onto the component: it is injective over $U_{j-1} \times \mathbb{A}^1 \times \mathbb{A}^{3(k-j)}$, by the same argument as **??** 5.6, applied at parameter j.

Proof of **??** *5.13.* By **??** 5.12, P_j's defining ideal is Y_2's own defining ideal run at parameter j – i.e. the component itself **is** Y_2 at parameter j, crossed with the free blocks $\overline{v}_{j+1}, \ldots, \overline{v}_k$. **??** 5.6's injectivity argument (over U_{j-1}, via **??** 4.2 at level $j-1$) applies verbatim at parameter j, and appending the free blocks q unchanged does not affect injectivity.

□

Remark 5.15: a reducible variety is not itself "rational" – component-by-component is what

Y_1 $V(I_{k-1}) \times \mathbb{A}^3$ (**??** 5.2) is **not** irreducible for $k \geq 4$ – it is a union of $k-2$ of the components above (**??** 5.11's inductive step) – so a single rational parametrization **of** Y_1 **as one object** is not meaningful in the usual sense (rationality is a property of irreducible varieties). What **is** meaningful, and what **??** 5.13 supplies, is a rational parametrization of each of Y_1's irreducible pieces individually – exactly the components inherited from $V(I_{k-1})$.

Example 5.16: the components, worked out at $k = 3$

Uniformly by k, since the components" **count** and **block pattern** are already known (**??** 5.11, **??** 5.12) and only their explicit generators change:

- **At** $k = 2$: I_2 is already prime (**??** 5.11's base case), so $V(I_2)$ has exactly $k - 1 = 1$ component – itself.
- **At** $k = 3$: exactly $k - 1 = 2$ components, both computed directly (Gröbner bases, not inferred from the pattern alone), matching $I_3 = P_2 \cap P_3$:
  - P_2, the "Y_1" piece (block pattern $\{1,2\}$, matching that the length-2 sub-bracket already vanishes with $\overline{v}_3$ free) – generated by exactly I_2's three generators from the first example, now inside R_3 with $\overline{v}_3$ simply not appearing:

$$v_{1,2}v_{2,3} - v_{1,3}v_{2,2}, \quad v_{1,3}v_{2,1} - v_{1,1}v_{2,3}, \quad v_{1,1}v_{2,2} - v_{1,2}v_{2,1}.$$

– P_3, the "Y_2" piece (block pattern $\{1,2,3\}$) – exactly the ideal already computed in the Y_2-defining-ideal example above:

$$2v_{2,2}v_{3,1} + 2v_{2,1}v_{3,2} + v_{2,3}v_{3,3}, \quad 2v_{1,2}v_{3,1} + 2v_{1,1}v_{3,2} + v_{1,3}v_{3,3},$$

$$2v_{1,2}v_{2,1}v_{3,2} - 2v_{1,1}v_{2,2}v_{3,2} - v_{1,3}v_{2,2}v_{3,3} + v_{1,2}v_{2,3}v_{3,3}.$$

- **At** $k = 4$: exactly $k - 1 = 3$ components, block patterns $\{1,2\}$, $\{1,2,3\}$, $\{1,2,3,4\}$:
  - The first two, P_2 and P_3, are inherited **verbatim** from $k = 3$ above, with $\overline{v}_4$ simply left free – not recomputed, per **??** 5.12" own "does not depend on which k" clause.
  - The third, P_4, is the new "Y_2" stratum – exactly the $k = 4$ row of the earlier Y_2-defining-ideal example (generators omitted here to avoid repeating that display; same growth pattern, quartic with up to 8 terms per generator).

6 Cohen-Macaulayness

Proposition 6.1: Eagon-Northcott / Hilbert-Burch, applied

Let M be a 2×3 matrix with entries in a polynomial ring R over a field (arbitrary entries, not necessarily independent indeterminates), and J the ideal of its 2×2 minors. If J has the **expected** codimension $3-2+1 = 2$, then J is a perfect ideal of grade 2 – $\mathrm{pd}(R/J) = 2$, hence R/J is Cohen-Macaulay.

Proof of **??** *5.15.* Bruns and Vetter BrunsVetter1988, Theorem (2.7), p. 13, gives exactly this for an **arbitrary** matrix, as needed since one row of our 2×3 matrix $(\tau(\overline{w}_{k-1}))$ is a fixed polynomial vector, not a row of independent indeterminates. (The original construction, Eagon and Northcott EagonNorthcott1962, is superseded here by Bruns-Vetter's more general treatment – confirmed closed-access via Unpaywall, not used directly.)

□

Remark 6.2: why the generator count above wasn't proven this way

It is tempting to expect **??** 5.15 to also hand over the generator count for free via Bruns-Vetter's Theorem (16.36), p. 217 ("The Theorem of Hilbert-Burch") – but that theorem, as actually stated, gives only $I = a \cdot I_m(f)$ for a nonzerodivisor a, not a generator count on its own: at expected codimension, an **arbitrary** 2×3 matrix's minors can be perfect and Cohen-Macaulay while one minor vanishes identically or is a scalar combination of the others (concretely: $\begin{pmatrix} x & y & 0 \\ 0 & 0 & 1 \end{pmatrix}$ over $\mathbb{F}[x,y]$ has minors $0, x, y$, height 2 as expected, $J = (x,y)$ perfect and Cohen-Macaulay with only **2** generators). This is exactly why **??** 4.6 was proven directly from linear independence instead of cited from this section's machinery.

Theorem 6.3: R_k/I_k is Cohen-Macaulay

For every $k \geq 2$: R_k/I_k is Cohen-Macaulay.

Proof of **??** *6.2.* By **??** 5.7, I_k has codimension 2, the expected value for the 2×3 matrix of **??** 3.3 presenting it. **??** 5.15 applies directly. (This uses **??** 5.7 only – not **??** 4.6's generator

count, not **??** 5.11's component count – genuinely independent of both.)

□

7 Radicality

Cohen-Macaulayness and the explicit component structure of §5 combine, via Serre's criterion, to give radicality – which then closes both the generator-count (§4) / Cohen-Macaulay (§6) story (making I_k itself, not just $\sqrt{I_k}$, the object with those properties) and upgrades §5's component count to a genuine primary decomposition.

Lemma 7.1: reduction via Serre

Recall Serre's condition S_n: a Noetherian ring A satisfies S_n if $\operatorname{depth}(A_P) \geq \min(n, \dim A_P)$ for every prime $P \subset A$ (Cohen-Macaulay means $\operatorname{depth}(A_P) = \dim A_P$ at **every** prime – the extremal case, $n = \infty$ in effect). Cohen-Macaulay rings satisfy S_n for every n: $\operatorname{depth}(A_P) = \dim A_P \geq \min(n, \dim A_P)$ trivially, whatever n is – this needs nothing beyond the two definitions just given, standard for Serre's conditions in general (see e.g. Bruns and Vetter BrunsVetter1988 or Eisenbud's *Commutative Algebra with a View Toward Algebraic Geometry*). In particular S_1. So, given **??** 6.2, radicality of I_k is equivalent to R_0: regularity of R_k/I_k at each of its minimal primes.

Lemma 7.2: Jacobian criterion: the rank bound

Let $S = \mathbb{F}[x_1, \ldots, x_n]$, $I = (f_1, \ldots, f_m)$, $A = S/I$ equidimensional of dimension $n - c$, and $x \in V(I)$ an $\mathbb{F}$-rational point with Jacobian $\operatorname{Jac}(x) \in \mathbb{F}^{m \times n}$. Then $\operatorname{rank}\ \operatorname{Jac}(x) \leq c$.

Proof of **??** *7.2.* $\dim_{\mathbb{F}} \mathfrak{m}/\mathfrak{m}^2 = n - \operatorname{rank}\ \operatorname{Jac}(x)$ always (linear algebra: the image of I in the cotangent space $\mathfrak{n}/\mathfrak{n}^2 \cong \mathbb{F}^n$ is exactly the span of the linear parts of generators of I, of dimension $\operatorname{rank}\ \operatorname{Jac}(x)$), and $\dim_{\mathbb{F}} \mathfrak{m}/\mathfrak{m}^2 \geq \dim A_{\mathfrak{m}}$ in any Noetherian local ring, so it suffices that $\dim A_{\mathfrak{m}} = n - c$ – which **??** 7.3 supplies.

□

Lemma 7.3: Jacobian criterion: regularity at rank $= c$

With hypotheses as in **??** 7.2: $\dim A_{\mathfrak{m}} = n - c$. Consequently, if $\operatorname{rank}\ \operatorname{Jac}(x) = c$, then $A_{\mathfrak{m}}$ is regular of dimension $n - c$; in particular x lies on a **unique** component of $V(I)$, whose local ring there is a field.

Proof of **??** *7.3.* $\dim A_{\mathfrak{m}} = \max\{\dim(A/P)_{\mathfrak{m}} : P \in \operatorname{Min}(A), P \subseteq \mathfrak{m}\}$, and each $A/P = S/\widetilde{P}$ is an affine domain over a field, for which $\dim\left(S/\widetilde{P}\right)_{\mathfrak{M}} = \dim S/\widetilde{P}$ at **every** maximal ideal $\mathfrak{M}$ – affine domains over a field are catenary and equicodimensional (Eisenbud, *Commutative Algebra*, Cor. 13.4; equivalently the Stacks Project, tag 00OS Stacks00OS – a dimension formula for finitely generated field extensions, holding over an arbitrary field, not needing a perfect-field hypothesis by accident). By equidimensionality every such $\dim S/\widetilde{P}$ equals $n - c$. If $\operatorname{rank}\ \operatorname{Jac}(x) = c$ then $\dim_{\mathbb{F}} \mathfrak{m}/\mathfrak{m}^2 = n - c = \dim A_{\mathfrak{m}}$ by **??** 7.2, the definition of regularity. A regular local ring is a domain, so $\mathfrak{m}$ contains a unique minimal prime of A, and the local ring at that prime is a further localization of a regular local ring at its own maximal ideal in dimension 0, hence a field.

□

Lemma 7.4: the Jacobian's own recursion

Write J_k for the $3 \times 3k$ Jacobian of $\overline{w}_k$, $[u]_\times$ for the skew matrix with $[u]_\times v = u \times v$, and T for the matrix of τ. Then $J_2 = \left[-[\overline{v}_2]_\times \,|\, [\overline{v}_1]_\times \right]$, and for $k \geq 3$,

$$J_k = \left[-[\overline{v}_k]_\times \, T J_{k-1} \,|\, [\tau(\overline{w}_{k-1})]_\times \right].$$

Proof of ?? 7.4. For the second block, $\overline{w}_k = \tau(\overline{w}_{k-1}) \times \overline{v}_k = [\tau(\overline{w}_{k-1})]_\times \, \overline{v}_k$ is linear in $\overline{v}_k$ with coefficient matrix free of $\overline{v}_k$. For the first, differentiating $\overline{w}_k = \tau(\overline{w}_{k-1}) \times \overline{v}_k$ with respect to any variable t among $\overline{v}_1, \ldots, \overline{v}_{k-1}$ gives $\partial_t \overline{w}_k = (T \partial_t \overline{w}_{k-1}) \times \overline{v}_k = -[\overline{v}_k]_\times \, T \partial_t \overline{w}_{k-1}$, using $a \times b = -[b]_\times a$; assembling columns gives the stated product.

□

Example 7.5: J_3, explicitly

With variables ordered $v_{1,1}, v_{1,2}, v_{1,3}, v_{2,1}, v_{2,2}, v_{2,3}, v_{3,1}, v_{3,2}, v_{3,3}$, the 3×9 Jacobian of $\overline{w}_3$ (the three generators of I_3 from the earlier example), column-blocked to match ?? 7.4's formula $J_3 = \left[-[\overline{v}_3]_\times \, T J_2 \,|\, [\tau(\overline{w}_2)]_\times \right]$ – shown here as the two blocks separately, since side by side the full 3×9 matrix does not fit the page width. First, $-[\overline{v}_3]_\times \, T J_2$ (3×6):

$$\begin{pmatrix} -2v_{2,2}v_{3,2} & 2v_{2,1}v_{3,2} + v_{2,3}v_{3,3} & -v_{2,2}v_{3,3} & 2v_{1,2}v_{3,2} & -2v_{1,1}v_{3,2} - v_{1,3}v_{3,3} & \\ 2v_{2,2}v_{3,1} + v_{2,3}v_{3,3} & -2v_{2,1}v_{3,1} & -v_{2,1}v_{3,3} & -2v_{1,2}v_{3,1} - v_{1,3}v_{3,3} & 2v_{1,1}v_{3,1} & \\ -v_{2,3}v_{3,2} & -v_{2,3}v_{3,1} & v_{2,2}v_{3,1} + v_{2,1}v_{3,2} & v_{1,3}v_{3,2} & v_{1,3}v_{3,1} & -v \end{pmatrix}$$

and then $[\tau(\overline{w}_2)]_\times$ (3×3, the skew matrix of $\tau(\overline{w}_2)$'s three coordinates from the earlier matrix example):

$$\begin{pmatrix} 0 & 2v_{1,2}v_{2,1} - 2v_{1,1}v_{2,2} & -v_{1,3}v_{2,2} + v_{1,2}v_{2,3} \\ -2v_{1,2}v_{2,1} + 2v_{1,1}v_{2,2} & 0 & -v_{1,3}v_{2,1} + v_{1,1}v_{2,3} \\ v_{1,3}v_{2,2} - v_{1,2}v_{2,3} & v_{1,3}v_{2,1} - v_{1,1}v_{2,3} & 0 \end{pmatrix}$$

Both blocks computed directly, not merely asserted from the recursive formula.

Remark 7.6: a false shortcut, caught and corrected

A naive reading of ?? 7.4 suggests appending $\overline{v}_k = 0$ preserves rank on an inherited component – this is false: the old-variable block is left-multiplied by $-[\overline{v}_k]_\times \, T$, not merely padded with zero rows. At the first inherited case ($k = 2 \to 3$), appending $\overline{v}_3 = 0$ actually drops the rank to 0. The correct witness (?? 7.8 below) instead picks $\overline{v}_k = e_i$ for a standard basis vector **not** lying in $T \cdot$ im $J_{k-1}(q)$ – a genuine choice, not a formality.

Lemma 7.7: rank-2 witness on the new component

For every $k \geq 2$, there is a point $x = (p, 0)$, p the ?? 4.2 non-degeneracy witness at level $k-1$, with $x \in V(I_k)$ and rank $J_{k(x)} = 2$.

Proof of ?? 7.5. By ?? 7.4, $J_{k(p,0)} = \left[0 \,|\, [\tau(\overline{w}_{k-1}(p))]_\times \right]$ since $-[0]_\times T J_{k-1}(p) = 0$; and $\tau(\overline{w}_{k-1}(p)) \neq 0$ since $\overline{w}_{k-1}(p) \neq 0$ (?? 4.2) and τ is invertible. For $u \neq 0$, $\ker [u]_\times = \{v : u \times v = 0\} = \mathbb{F}u$ (a cross product vanishes exactly on linearly dependent pairs), so $[\tau(\overline{w}_{k-1}(p))]_\times$ has rank 2, giving rank $J_{k(p,0)} = 2$.

□

Lemma 7.8: rank-2 witness on an inherited component

For every $k \geq 3$ and every component Z of $V(I_{k-1})$ with a rank-2 witness $q \in Z$ (?? 7.5, or this lemma applied at $k-1$), there is $x = (q, e_i) \in Z \times \mathbb{A}^3 \subset V(I_k)$ with rank $J_{k(x)} = 2$.

Proof of ?? 7.8. Let $W\ T \cdot$ im $J_{k-1}(q)$, a 2-dimensional subspace of $\mathbb{F}^3$ since T is invertible and rank $J_{k-1}(q) = 2$. Since e_1, e_2, e_3 span $\mathbb{F}^3$ and $W \neq \mathbb{F}^3$, some $e_i \notin W$; fix such an i. Since $q \in V(I_{k-1})$, $\overline{w}_{k-1}(q) = 0$, so ?? 7.4 gives $J_{k(q,e_i)} = \left[-[e_i]_\times TJ_{k-1}(q) | 0\right]$, of rank $\dim([e_i]_\times W)$ since $TJ_{k-1}(q)$ has image exactly W. As $\ker [e_i]_\times = \mathbb{F}e_i$ and $W \cap \mathbb{F}e_i = 0$ (else $e_i \in W$), $[e_i]_\times$ is injective on W, so $\dim([e_i]_\times W) = 2$.

□

Theorem 7.9: I_k is radical

For every $k \geq 2$: I_k is radical.

Proof of ?? 7.9. By induction on k (mirroring ??'s own induction), ?? 7.5 and ?? 7.8 exhibit a rank-2-Jacobian point on **every** one of the $k-1$ components of ?? 5.11. ?? 7.3 then gives R_0 at every one (and, by its "unique component" clause, confirms each witness lies only on its intended component – an independent check on ??'s count, not merely assumed); ?? 7.1 gives S_1, using ?? 6.2's Cohen-Macaulayness. Serre's criterion gives radicality.

□

Verified independently in Sage at every $k = 2, \ldots, 6$: the Jacobian recursion (?? 7.4) checked as a polynomial identity, not just at points, and each witness confirmed to lie on **exactly** its intended component via an independently-computed primary decomposition (code/verify_jacobian_rank_all_co

8 The Primary Decomposition Theorem

Every ingredient is now in hand for the fully general statement – component count (§5) and radicality (§7) together, neither sufficient alone:

Theorem 8.1: Primary Decomposition Theorem

For every $k \geq 2$, over any field of characteristic $\neq 2$: I_k has exactly $k-1$ irreducible, radical, primary components, namely $P_2, \ldots, P_k$ of ?? 5.12, and

$$I_k = P_2 \cap \cdots \cap P_k$$

is an irredundant primary decomposition into $k-1$ primes.

Proof of ?? 8.1. Spelled out in full, combining the two independent halves proven above rather than left as a one-line conjunction. By ?? 7.9, I_k is radical, i.e. $I_k = \sqrt{I_k}$. A standard fact about radical ideals in a Noetherian ring: a radical ideal equals the intersection of its own minimal primes, $I_k = \cap_{P \in \text{Min}(I_k)} P$ – since $\sqrt{I_k}$ is, by definition, the intersection of **all** primes containing I_k, and the minimal ones already suffice (every non-minimal prime containing I_k contains some minimal one, hence is redundant in the intersection). ?? 5.11 identifies Min (I_k) explicitly: exactly $k-1$ primes, one per block pattern $j = 2, \ldots, k$, named $P_2, \ldots, P_k$ by ?? 5.12. Substituting gives the displayed equation. Each P_j, being prime, is trivially its own primary-decomposition component (P_j-primary with associated prime P_j itself) – no separate argument for "primary" beyond "prime" is needed, and primality of each P_j is exactly ?? 5.3 (applied at the appropriate parameter, via ?? 7.9's own induction). Irredundancy – no P_j can

be omitted, i.e. no P_j contains $\cap_{i \neq j} P_i$ – is exactly **??** 5.11's own no-containment conclusion (Proposition 17 and Proposition 18, applied at every level of that theorem's induction), not a fact needing any further proof here.

□

Remark 8.2: summary

?? 5.7 (codimension 2), **??** 4.6 (3 minimal generators), **??** 6.2 (Cohen-Macaulay), and **??** 7.9 (radical) together say exactly what a single bundled "I_k has codimension 2, is radical, has exactly 3 minimal generators, and is Cohen-Macaulay" statement would – now as four independently-proven results, each established at the point in this article where its actual dependencies (and no others) were available, rather than one claim proven in parts across several sections.

9 Summary

For every $k \geq 2$, over any field $\mathbb{F}$ of characteristic $\neq 2$:

- I_k has codimension 2 (**??** 5.7), presented as the ideal of 2×2 minors of an explicit 2×3 matrix (**??** 3.3) – a genuine determinantal ideal, not merely determinantal-looking.
- I_k has exactly 3 minimal generators (**??** 4.6), proven directly from linear independence, independent of the determinantal presentation.
- R_k/I_k is Cohen-Macaulay (**??** 6.2), from the determinantal presentation and classical Eagon-Northcott/Hilbert-Burch theory (**??** 5.15).
- I_k is radical (**??** 7.9), via Serre's criterion: Cohen-Macaulayness gives S_1 (**??** 7.1), and an explicit rank-2 Jacobian witness on every component gives R_0 (**??** 7.5, **??** 7.8).
- $V(I_k)$ has exactly $k-1$ irreducible components (**??** 5.11), each rational – in fact birational to an explicit affine parameter space (**??** 5.13) – and these assemble, via the two facts above, into the irredundant primary decomposition of the Primary Decomposition Theorem (**??** 8.1): $I_k = P_2 \cap \cdots \cap P_k$.

Nothing above is conjectural or checked only at small k: every statement is proven for **every** $k \geq 2$, over an **arbitrary** field of the stated characteristic. The one running computational thread – Sage verification of the determinantal identity, the Jacobian recursion, and the witness points, at $k = 2, \ldots, 6$ – corroborates the general proofs; it is not a substitute for them anywhere in this article.

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